e) Asking whether the linear system corresponding to an augmented matrix $[a_1\ a_2\ a_3\ b]$ has a solution amounts to asking whether b is in Span$\{a_1, a_2, a_3\}$.

$\boxed{\text{Solution:}}$ This statement is True. We justify this by whats given on Page 30 under the second grey box. We state: Asking whether a vector b is in Span$\{v_1, \ldots, v_p\}$ amounts to asking whether the vector equation $\boxed{x_1v_1 + x_2v_2 + \ldots + x_pv_p = b}$ has a solution, or, equivalently, asking whether the linear system with augmented matrix $[v_1 \ldots v_p\ b]$ has a solution.

25) Let $A = \begin{bmatrix} 1 & 0 & -4 \\ 0 & 3 & -2 \\ -2 & 6 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 4 \\ 1 \\ -4 \end{bmatrix}$. Denote the columns of A by a_1, a_2, a_3 and let $W = \text{Span}\{a_1, a_2, a_3\}$

a) Is b in $\{a_1, a_2, a_3\}$? How many vectors are in $\{a_1, a_2, a_3\}$?

$\boxed{\text{Solution:}}$ It's easy to see that the column b does not match with any column of A. $\therefore$ b is not in $\{a_1, a_2, a_3\}$.

There are 3 vectors in $\{a_1, a_2, a_3\}$.

b) Is b in W? How many vectors are in W?

$\boxed{\text{Solution:}}$ By defn. of Span, b is in W if b is a linear combination of the columns a_1, a_2 and a_3, i.e. $\boxed{x_1a_1 + x_2a_2 + x_3a_3 = b}$ for some $x_1, x_2, x_3 \in \mathbb{R}$.

This is equivalent to claiming that the augmented matrix $[a_1\ a_2\ a_3\ b]$ has a solution set.

$$\begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ -2 & 6 & 3 & -4 \end{bmatrix} \xrightarrow[+2\times \text{Row}1]{\text{Row}3 = \text{Row}3} \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ 0 & 6 & -5 & 4 \end{bmatrix} \xrightarrow[\text{Row}3 - 2\times \text{Row}2]{\text{Row}3 =} \begin{bmatrix} 1 & 0 & -4 & 4 \\ 0 & 3 & -2 & 1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$